

FIELD GUIDE FG-001

ONE-PAGER: LONG CONTENT CONDENSATION

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Comprehensive WAFT Guide ## Introduction WAFT (Wave Agent Framework & Tools) is a Python meta-framework for directed evolution of self-modifying AI agents. It provides a scientific instrument for studying the physics of artificial cognition. ## The Three Pillars #### 1. The Substrate (Code as DNA) Every agent has a genome - a SHA-256 hash of its code, configuration, and prompts. Mutations are changes to this genome. Evolution happens through hot-swapping genomes. #### 2. The Physics (Scint System) Reality fracture detection serves as natural selection. Errors are classified as: - SYNTAX_TEAR: Formatting errors - LOGIC_FRACTURE: Mathematical/logical errors - SAFETY_VOID: Harmful outputs - HALLUCINATION: Fabricated information Fitness = (Stability × 0.4) + (Efficiency × 0.3) + (Safety × 0.3) Agents with fitness < 0.5 are considered DEAD. #### 3. The Flight Recorder Telemetry tracks: - Genome ID - Parent ID - Generation number - Event type - Context - Fitness score This enables family tree reconstruction and scientific analysis. ## Quick Start ```bash uv tool install waft waft new my_laboratory cd my_laboratory waft verify ``` ## Evolutionary Cycle 1. **Spawn**: Create variants 2. **Evaluate**: Test fitness 3. **Evolve**: Adopt best variant 4. **Record**: Track all changes ## Key Features - Self-modification capabilities - Evolutionary tracking system - Fitness testing framework - Scientific data collection - Gamification elements - Epistemic tracking ## Essential Commands **Project Management:** - `waft new` - Create new project - `waft verify` - Check system health - `waft status` - Show current state **Evolution:** - `waft spawn --agent` - Create variant - `waft eval --agent` - Evaluate fitness - `waft evolve --agent` - Adopt best variant **Documentation:** - `waft-docs` - Generate documentation - `waft-status` - Generate status reports ## Goal Observe the "God-Head" agent emerge from thousands of generations. This produces data for "The Physics of Artificial Cognition" research. ## Version - Current: 0.5.0 - Python: 3.10+ - Package Manager: uv

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